THE LANGUAGE OF

# MATHEMATICS

## General notation

| Symbol | Meaning | Example |
|---|---|---|
| $=$ | Equal to | $2+3=5$ |
| $\neq$ | Not equal to | $2 \neq 3$ |
| $\approx$ | Approximately equal to | $\pi \approx 3.14$ |
| $<,\ >$ | Less than; greater than | $2<3; \quad 5>4$ |
| $\leq,\ \geq$ | Less than or equal to; greater than or equal to | $2 \leq 2$<br>$5 \geq 4$ |
| $\pm$ | Plus or minus | $3 \pm 1 = 4$ or $2$ |
| $\lvert a \rvert$ | Absolute value | $\lvert -5 \rvert = 5$ |
| $\sqrt{a}$ | Square root | $\sqrt{9}=3$ |

## Sets and number systems

| Symbol | Meaning | Example |
|---|---|---|
| $\{a,b,c\}$ | Set with three elements | $\{2,4,6\}$ |
| $\varnothing$ | Empty set | $\{1\} \cap \{2\} = \varnothing$ |
| $\in,\ \notin$ | Is an element of; is not an element of | $2 \in \{1,2\}$<br>$3 \notin \{1,2\}$ |
| $\subseteq$ | Is a subset of | $\{1\} \subseteq \{1,2\}$ |
| $\cup,\ \cap$ | Union; intersection | $\{1\} \cup \{2\} = \{1,2\}$<br>$\{1,2\} \cap \{2\} = \{2\}$ |
| $\mathbb{N},\ \mathbb{Z}$ | Natural numbers; integers | $3 \in \mathbb{N}; \quad -2 \in \mathbb{Z}$ |
| $\mathbb{Q},\ \mathbb{R}$ | Rational numbers; real numbers | $\frac{1}{2} \in \mathbb{Q}; \quad \sqrt{2} \in \mathbb{R}$ |
| $(a,b),\ [a,b]$ | Open interval; closed interval | $1 \notin (1,3)$<br>$1 \in [1,3]$ |

## Logic

| Symbol | Meaning | Example |
|---|---|---|
| $\Rightarrow$ | Implies | $x>2 \Rightarrow x>1$ |
| $\Leftrightarrow$ | If and only if | $x+1=3 \Leftrightarrow x=2$ |
| $\forall$ | For every | $\forall x \in \mathbb{R},\ x^2 \geq 0$ |
| $\exists$ | There exists | $\exists x \in \mathbb{Z}:\ x+2=0$ |
| $\neg$ | Logical negation | $\neg(2>3)$ is true |

## Summation notation

| Symbol | Meaning | Example |
|---|---|---|
| $\sum_{i=1}^{n} a_i$ | Add $a_i$ for each integer $i$ from 1 to $n$. | $\sum_{i=1}^{3} 2i = 2+4+6$ |

In $\sum_{i=1}^{n} a_i$, $i$ is the **index**, 1 is the **lower bound**, $n$ is the **upper bound**, and $a_i$ is the **expression being summed**.

### Worked examples

$$\sum_{i=1}^{4} i = 1+2+3+4 = 10$$

$$\sum_{k=1}^{3} k^2 = 1+4+9 = 14$$

Symbols acquire meaning from their mathematical context.

## Probability and events

| Symbol | Meaning | Example |
|---|---|---|
| $\Omega$ | Sample space: all possible outcomes | $\Omega = \{1,2,3,4,5,6\}$ |
| $\omega$ | An individual outcome | $\omega = 3 \in \Omega$ |
| $A=\{1,3,5\}$ | Event shown as a set of outcomes | $A=\{1,3,5\}$: odd roll |
| $A \subseteq \Omega$ | An event is a subset of the sample space | $\{1,3,5\} \subseteq \Omega$ |
| $P(A)$ | Probability of event $A$ | $P(A)=\frac{3}{6}=\frac{1}{2}$ |
| $A^c$ | Complement of event $A$ | $A^c=\{2,4,6\}$ |
| $A \cup B$ | Event $A$ or $B$ (or both) | $B=\{1,2\}$<br>$A \cup B = \{1,2,3,5\}$ |
| $A \cap B$ | Both events $A$ and $B$ | $A \cap B = \{1\}$ |
| $P(A \mid B)$ | Probability of $A$ given $B$ | $P(A \mid B)=\frac{1}{2}$ |

Probability examples use a fair six-sided die, $A=\{1,3,5\}$ and $B=\{1,2\}$.

## Function notation

| Symbol | Meaning | Example |
|---|---|---|
| $f$ | Name of a function | $f(x)=x^2$ |
| $f(x)$ | Value of $f$ at input $x$ | $f(3)=3^2=9$ |
| $f: A \to B$ | Function from set $A$ to set $B$ | $f: \mathbb{R} \to \mathbb{R},\ f(x)=x^2$ |
| $x \mapsto f(x)$ | Input-to-output mapping | $x \mapsto x^2; \quad 3 \mapsto 9$ |
| $\mathrm{dom}(f)$ | Domain of a function | $\mathrm{dom}(f)=\mathbb{R}$ |

## Letters and Greek symbols

| Symbol | Meaning | Example |
|---|---|---|
| $a,b,c,\ldots$ | Common lowercase Latin letters | $a=2,\ b=3,\ c=a+b$ |
| $A,B,C,\ldots$ | Common uppercase Latin letters | $A=\{1\},\ B=\{2\}$<br>$C=A \cup B$ |
| $x,y,z$ | Other common Latin letters | $y=2x, \quad z=x+y$ |
| $\alpha,\ \beta,\ \gamma$ | Alpha; beta; gamma | Triangle angles:<br>$\alpha+\beta+\gamma=180°$ |
| $\delta,\ \Delta$ | Delta; uppercase delta | $\delta=0.1; \quad \Delta x = x_2 - x_1$ |
| $\theta,\ \Theta$ | Theta; uppercase theta | $\theta=30°; \quad \Theta=2\theta$ |
| $\lambda,\ \Lambda$ | Lambda; uppercase lambda | $\lambda=2; \quad \Lambda=\{1,2\}$ |
| $\mu,\ \sigma,\ \Sigma$ | Mu; sigma; uppercase sigma | $\mu=\frac{3+7}{2}=5$<br>$\sigma=\sqrt{4}=2$<br>$\sum_{i=1}^{3} i = 6$ |
| $\pi,\ \Pi$ | Pi; uppercase pi | $\pi \approx 3.14$<br>$\prod_{j=1}^{3} j = 6$ |
| $\phi,\ \Phi$ | Phi; uppercase phi | $\phi=\frac{1+\sqrt{5}}{2}; \quad \Phi=\phi^2$ |
| $\omega,\ \Omega$ | Omega; uppercase omega | $\Omega=\{H,T\}; \quad \omega=H$ |